

Title: Tower of Hanoi - Detailed Explanation (Classic & Modified Versions)

■ Classic Tower of Hanoi

Problem

We have 3 rods (A, B, C) and n disks (different sizes). Move all disks from A to C following:

1. Only one disk at a time.
2. Never place a larger disk on a smaller one.

Recursive Strategy

To move n disks from A → C using B:

1. Move n-1 disks from A → B
2. Move the nth disk from A → C
3. Move n-1 disks from B → C

Code Logic (Classic)

```
...  
public static void tow(int n, char src, char dest, char aux){  
    if(n==0) return;  
    tow(n-1,src,aux,dest);  
    System.out.println("Move "+n+" from "+src+" to "+dest);  
    tow(n-1,aux,dest,src);  
}  
...
```

The direct move line is:

```
...  
System.out.println("Move " + n + " from " + src + " to " + dest);  
...
```

This is where the disk physically moves from source to destination.

Number of Moves

Formula = $(2^n) - 1 \rightarrow$ printed as ``(1 << n) - 1``.

■■ Modified Tower of Hanoi (No Direct A→C Move)

Concept

In this variation, a disk ****cannot move directly from source to destination****. It must always go through the auxiliary rod.

Base Case

If only one disk:

- Move from src → aux
- Then from aux → dest

Recursive Steps

```
...  
TowerOfHanoi(n - 1, src, aux, dest);  
System.out.println("Move " + n + " from " + src + " to " + aux);  
TowerOfHanoi(n - 1, dest, aux, src);  
System.out.println("Move " + n + " from " + aux + " to " + dest);  
TowerOfHanoi(n - 1, src, aux, dest);
```

...

This ensures every disk passes through the auxiliary rod twice.

Move Count Formula

...

$\text{COUNT}(n) = 3 * \text{COUNT}(n - 1) + 2$

$\text{COUNT}(1) = 2$

...

n	COUNT(n)
---	----------

1	2
---	---

2	8
---	---

3	26
---	----

4	80
---	----

Differences

Feature	Classic	Modified
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Direct Move	Allowed (src→dest)	Not Allowed
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Formula	$2^n - 1$	$3 * \text{COUNT}(n-1) + 2$
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Base Move Count	1	2
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Recursion Depth	Moderate	High
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■ In Summary

- The classic version allows direct movement → simple $2^n - 1$ moves.
- The modified version enforces extra constraints → far more complex recursion.
- Both show the **power of recursion**: breaking big problems into smaller copies of themselves.

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